

BOTEV VRAH, BulgariaWMO: **156270**Lat: **42.7170N**Lon: **24.9170E**Elev: **7838**StdP: **10.98**Time Zone: **2.00 (EUE)**Period: **94-13**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-3.1	0.0	-11.6	3.9	7.1	-7.4	4.9	8.2	77.0	14.9	76.6	15.1	27.5	320	1.073

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.3	60.9	50.5	58.3	49.7	56.1	48.9	53.5	57.1	51.9	55.7	50.4	54.1	11.3	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
52.1	77.7	55.0	50.4	72.8	53.2	48.9	68.8	51.8	25.5	57.2	24.4	55.9	23.4	54.3	61.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 76.2	63.1	53.7	-6.8	65.7	4.9	2.5	-10.3	67.5	-13.2	69.0	-16.0	70.4	-19.6	72.3		
(5)			-7.4	56.8	4.9	1.8	-11.0	58.1	-13.8	59.1	-16.6	60.1	-20.1	61.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	32.4	17.8	17.5	21.2	28.5	37.5	43.6	47.4	48.2	41.4	35.5	28.5	21.0	(6)
(7)		DBStd	13.29	8.70	8.88	8.41	6.59	6.28	6.53	5.63	5.37	6.38	7.90	9.53	8.89	(7)
(8)		HDD50	6519	999	910	891	646	387	209	116	99	266	450	647	899	(8)
(9)		HDD65	11886	1464	1330	1356	1096	851	642	545	520	709	913	1096	1364	(9)
(10)		CDD50	108	0	0	0	0	1	17	37	44	7	2	0	0	(10)
(11)		CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)
(13)		CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	20.0	23.4	24.0	24.6	21.5	18.0	17.4	16.1	14.2	16.3	18.8	22.2	23.1	(14)
(15)	Precipitation	PrecAvg	31.7	2.2	2.0	2.2	2.6	4.0	4.1	4.0	2.6	2.1	1.9	1.5	2.5	(15)
(16)		PrecMax	48.7	5.3	3.5	5.2	5.6	7.9	8.4	10.6	7.4	6.6	4.9	3.9	5.0	(16)
(17)		PrecMin	23.3	0.6	0.7	1.1	0.9	0.6	0.7	0.3	0.0	0.2	0.2	0.2	0.7	(17)
(18)		PrecStd	6.4	1.1	0.7	1.0	1.2	2.0	1.9	2.8	1.9	1.5	1.1	0.8	0.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.7	37.9	43.5	46.8	55.0	62.0	65.7	64.4	58.2	55.7	50.0	41.6	(19)
(20)			MCWB	31.2	28.4	32.3	39.7	44.8	50.6	51.9	53.0	47.4	43.7	37.0	32.9	(20)
(21)		2%	DB	33.9	33.2	38.3	43.3	50.9	57.4	61.1	60.9	55.0	51.0	46.8	37.6	(21)
(22)			MCWB	27.1	27.8	31.3	38.0	43.4	49.6	50.1	50.8	46.0	41.6	35.6	31.0	(22)
(23)		5%	DB	31.0	30.6	34.9	39.3	48.0	55.1	58.0	58.6	52.6	48.4	43.5	35.1	(23)
(24)			MCWB	27.1	26.9	29.7	35.3	42.6	48.7	50.1	49.8	44.9	40.1	33.7	30.1	(24)
(25)		10%	DB	28.5	28.0	32.2	36.3	45.8	52.6	55.3	56.0	50.0	45.4	40.3	32.3	(25)
(26)			MCWB	25.7	25.7	29.1	33.4	41.8	47.9	49.3	49.2	44.6	38.7	32.7	28.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.3	31.4	36.2	42.1	47.8	54.3	55.5	56.3	50.7	46.8	41.9	37.4	(27)
(28)			MADB	34.3	33.6	40.2	45.1	51.3	57.1	59.4	60.8	55.1	50.2	46.1	38.6	(28)
(29)		2%	WB	29.9	29.2	32.7	38.7	45.2	52.2	53.2	53.7	48.4	44.1	38.9	33.9	(29)
(30)			MADB	32.2	31.7	35.6	42.3	48.1	55.0	57.3	57.5	52.6	48.1	43.1	35.6	(30)
(31)		5%	WB	27.9	27.5	30.9	35.9	43.7	50.5	51.7	51.9	46.8	42.6	36.2	31.2	(31)
(32)			MADB	29.8	29.6	33.3	38.2	46.7	53.4	55.6	56.1	50.4	46.0	38.8	33.5	(32)
(33)		10%	WB	26.1	25.9	29.3	33.9	42.3	48.6	50.3	50.2	45.5	40.9	34.4	29.0	(33)
(34)			MADB	27.7	27.9	31.8	36.0	45.1	51.7	53.7	54.3	48.6	43.5	38.5	30.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.5	7.8	7.8	6.7	6.5	6.9	7.3	7.3	6.8	7.1	7.4	7.5	(35)
(36)			MCDBR	9.1	9.2	8.5	8.0	8.0	9.5	10.6	9.4	9.7	8.5	7.8	8.5	(36)
(37)			MCWBR	6.8	7.6	6.7	7.1	7.7	8.9	8.3	8.5	7.9	7.1	5.9	6.6	(37)
(38)		5% WB	MCDBR	7.8	7.8	7.7	7.5	7.3	7.8	8.9	8.7	7.8	6.9	6.5	7.9	(38)
(39)			MCWBR	6.8	7.5	6.9	7.2	7.4	8.9	8.7	8.7	7.3	7.0	6.6	7.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.247	0.268	0.299	0.337	0.327	0.329	0.331	0.337	0.318	0.291	0.269	0.251	(40)	
(41)		taud	2.465	2.445	2.422	2.314	2.386	2.390	2.379	2.360	2.413	2.494	2.531	2.491	(41)	
(42)		Ebn at Noon	291	300	301	295	300	298	296	291	290	287	280	280	(42)	
(43)		Edh at Noon	23	28	32	39	37	37	37	37	32	26	21	20	(43)	
(44)	All-Sky Solar Radiation	RadAvg	502	756	1089	1453	1776	1953	2022	1827	1353	894	561	427	(44)	
(45)		RadStd	72	97	98	159	143	149	149	128	103	101	61	58	(45)	